

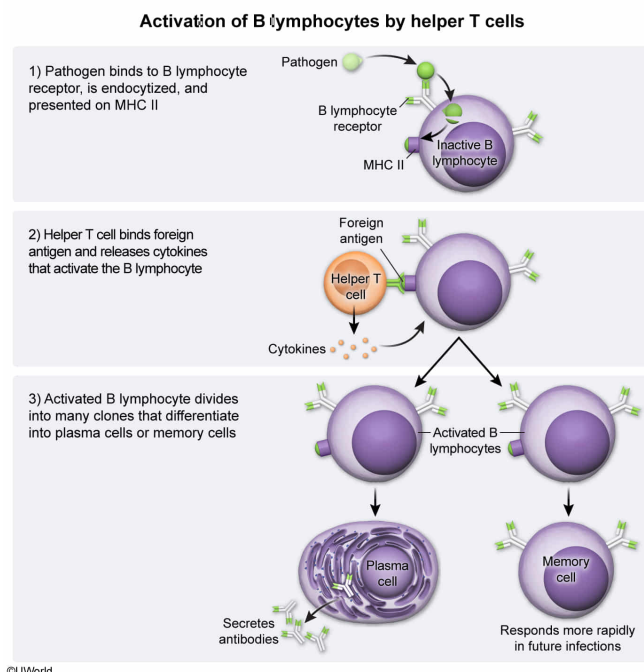
If a viral antigen were to bind to the receptors on the surface of a B lymphocyte, which of the following immune responses would NOT occur?

- ☐ A. Division of the B lymphocyte into cells that differentiate to secrete antibodies [14%]
- ☐ B. Interaction of the B lymphocyte with other immune cells that stimulate B lymphocyte proliferation [10%]
- ✓ ☒ C. Secretion of toxins by the B lymphocyte that destroy nearby virus-infected cells [66%]
- ☐ D. Proliferation of cells that can recognize the antigen more rapidly in the event of a future infection [8%]

Correct

66% answered correctly

Explanation:



The **immune response** is divided into **innate immunity** and **adaptive immunity**. The **innate immune system** comprises the skin and mucous membranes as well as specialized proteins and cells that rapidly and nonspecifically recognize and destroy antigens (ie, molecules targeted by the immune system). In contrast, the **adaptive immune system** comprises cells that mount specialized immune responses based on the learned recognition of specific antigens.

Adaptive immunity is further subdivided into **cell-mediated immunity** and

humoral immunity. Specifically, humoral immunity is driven by **B lymphocytes** that secrete **antibodies**, specialized proteins that recognize specific foreign antigens by binding to epitopes (eg, regions on viral or bacterial antigens to which antibodies bind). These antibodies circulate throughout the blood and lymph, binding foreign antigens and marking them for destruction by other immune cells.

Mature B lymphocytes are *inactive* until their receptors bind a specific foreign antigen. On activation by antigen binding, B lymphocytes can endocytose (engulf) the antigen, breaking it down into smaller fragments. Within the B lymphocyte, specialized antigen-binding molecules called **major histocompatibility class II (MHC II) proteins** bind the antigen fragments. Although all nucleated cells can display antigens on MHC I proteins, typically specialized ("professional") antigen-presenting cells display antigens on MHC II proteins.

MHC-antigen complexes are transported to the surface of the B lymphocyte and integrate into the cell membrane, allowing antigen fragments to be displayed outside the cell and presented to other immune cells known as **helper T cells**. On binding antigens presented by B lymphocytes, helper T cells **release** cytokines (signaling molecules) that stimulate B lymphocyte proliferation (**Choice B**). This produces many identical B lymphocytes that may differentiate into either of the following cell types:

- **Short-lived plasma cells** *secrete antibodies* during the immediate immune response (**Choice A**).
- **Long-lived memory cells** remain in **lymphoid tissues** for a long time and can **recognize the antigen more rapidly** in the event of a **future infection** (**Choice D**).

The question asks for an immune response that would not occur in this scenario. **Natural killer and cytotoxic T cells**, *not* B lymphocytes, respond to antigens by releasing toxins that induce apoptosis (cell death) in nearby infected cells.

Educational objective:

Humoral immunity is an antibody-mediated immune response. B lymphocytes bind foreign antigens, become activated, and divide into many identical cells, some of which differentiate into short-lived plasma cells that secrete antibodies during the immediate immune response. The remaining cells differentiate into

long-lived memory cells, which can respond more rapidly to future infections.

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